

90-Day Patient Lifestyle & Glucometer Model

Rules and Definitions

1. Morning Running Data (90 Days)

- The dataset covers 90 consecutive days, labelled Day 1 through Day 90.
- Each day records the kilometres run in the morning by a person.

2. Morning Walking Data

- A walking distance is recorded alongside the running distance for each of the 90 days.

3. Patients and Day 0 Glucometer Readings

- There are 40 patients, each with a unique name and an ID from P001 to P040.
- Each patient has an initial (Day 0) glucometer reading taken before the routine begins.

4. Medication Adherence Grid

- For each patient, medication is tracked over 90 days with three doses per day: Morning, Evening, and Night.

5. Meal Adherence Grid

- Meals are tracked in the same structure: 90 days x 3 (Morning, Evening, Night) = 270 meal slots per patient.

6. Running Schedule Adherence (%)

- Each patient has a running-schedule adherence percentage

7. Running Factor

For the running factor, the days are split into five bands.

Band	Definition	
A	Below 2 km	
B	Between 2 km and Ref 1	
C	Between Ref 1 and Ref 2	
D	Between Ref 2 and Ref 3	
E	Above Ref 3	

Each of the 90 days falls into exactly one band based on the running distance:

Using the five band day-counts A, B, C, D, E above, the running factor is chosen by the patient's running-schedule percentage:

If running % is below 85:

$$\text{Running factor} = A \times 0.85 + B \times 0.75 + C \times 0.65 + D \times 0.55 + E \times 0.65$$

If running % is between 85 and 90:

$$\text{Running factor} = A \times 0.85 + B \times 0.85 + C \times 0.65 + D \times 0.55 + E \times 0.65$$

If running % is between 90 and 95:

$$\text{Running factor} = A \times 0.85 + B \times 0.85 + C \times 0.85 + D \times 0.55 + E \times 0.65$$

If running % is above 95:

$$\text{Running factor} = A \times 0.85 + B \times 0.85 + C \times 0.85 + D \times 0.75 + E \times 0.65$$

8. Final Glucometer Reading after 90 Days

The final reading for each patient is computed as:

$$\begin{aligned} \text{Final reading} = & \text{Initial reading} - (\text{number of times medicine taken out of 270} \\ & \text{scheduled})/2 - (\text{number of times meal taken out of 270 scheduled})/4 + (\text{number of times} \\ & \text{only medicine skipped out of 270 scheduled}) + (\text{number of times only meal skipped out} \\ & \text{of 270 scheduled}) + 3 \times (\text{number of times both medicine and meal skipped out of 270} \\ & \text{scheduled}) - \text{running factor} \end{aligned}$$

Where, for each patient:

- Initial reading is the Day 0 glucometer value (Section 3).
- Running factor is determined as in Section 7.